

CS-3009 Software Engineering

Group Members

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Final Report

Lost and Found Management System

1. Introduction

The Lost and Found Management System aims to streamline the process of reporting, tracking, and recovering lost items within a university, corporate office, or public space. The system provides an efficient and user-friendly interface for users to log missing items, report found objects, and facilitate communication between the finder and the owner.

2. Project Vision

Our goal is to build a centralized, digital platform where users can report lost and found items efficiently, reducing manual work and increasing item recovery success rates. The system will ensure transparency, security, and ease of use, fostering a trustworthy community-driven approach to lost and found management.

3. Intended Use of the System

1. **Students & Employees** – To report and claim lost items.
2. **Security/Admin Staff** – To manage reported items and validate claims.
3. **Public Users** – To report found objects in public places.

Stakeholders & Their Needs

Stakeholders	Needs
Users	Quick item reporting, search, claim process.
Security/Admins	
Developers	Verification of lost and found items.
University/Organization	System scalability and maintenance. Reduce lost property cases, improve campus management.

Features & Overall Functionality

System Capabilities:

Users can report a lost item with details (date, location, description, image).
Users can report a found item and upload relevant information.
Automated matching algorithm to connect reported lost items with found items.
Admin verification system to prevent fraudulent claims.
Users receive notifications when their item is matched.
Search & filter options to browse lost/found items by category, date, and location.
Secure user authentication and access control.

How the System Helps Users?

Reduces time and effort required to find lost items.
Provides real-time notifications for matched items.
Ensures verified claims to avoid false ownership disputes.
Creates a centralized database, reducing redundant reporting.

Functional Requirements

1. User Registration & Authentication

- Users can register via email, phone number, or third-party authentication (Google).
- Passwords must be encrypted using secure hashing (e.g., bcrypt).

2. Lost Item Reporting

- Users can fill out a form with details (item name, category, description, location, date, and image).

3. Found Item Reporting

- Users can submit information about found items with optional image uploads.

4. Smart Matching

- The system automatically matches lost and found items using keywords, metadata, and image similarity algorithms.

5. Search and Filtering

- Users can search for lost/found items based on:
 - Category
 - Location
 - Date
 - Keywords

6. Claim Process

- Users can submit a claim on a found item, providing proof of ownership.
- Admins can approve or reject claims.

7. Admin Verification

- Admins can review submitted claims, reported items, and resolve disputes.

8. Image Upload and Preview

- Users can upload item images in formats like JPG, PNG.
- Preview functionality is available before submission.

9. Messaging

- Secure in-app messaging between finder and claimant with optional masked contacts.

10. Status Updates

- Users can mark items as "Still Lost," "Claimed," or "Resolved."

11. Report Fraudulent Listings

- Users can report suspicious listings for admin review.

12. Notifications

- Notifications are sent via:
 - Email
 - SMS
 - In-app alerts
 -

13. Location-Based Search

- Users can search for items based on their current or selected location.

14. Admin Dashboard

- Admins can:
 - View system metrics (e.g., active listings, number of resolved cases).
 - Moderate listings and disputes.

15. Social Media Sharing

- Users can share their listings on platforms like Facebook, Twitter, WhatsApp with safe, privacy-protected links.

Non-Functional Requirements (NFRs)

Category	Requirements
Performance	System should support at least 500 concurrent users without notable latency (<2s page load time).
Security	Use JWT tokens for session management. All passwords must be hashed (bcrypt/argon2). Data encryption for sensitive information (e.g., user contacts).
Usability	Clean and intuitive UI with clear navigation for both mobile and desktop users. Time to complete an item report should not exceed 2 minutes for average users.
Scalability	System should support future integration with external databases (e.g., university records) and third-party notification systems (e.g., Twilio, Firebase Cloud Messaging).
Reliability	System should maintain 99.5% uptime monthly. Recovery procedures should be in place in case of failure (database backup daily).
Maintainability	Codebase must follow modular architecture (MVC or MVVM), allowing easy updates and bug fixes.
Portability	Web-based with responsive design; future Android/iOS app compatibility planned.
Compliance	Compliant with GDPR standards for user data privacy where applicable.

User Stories Brief Descriptions

US1: Post Lost Item

As a user, report a lost item by providing details (description, location, date, image) to help others find it.

US2: Post Found Item

As a user, report a found item by providing details (description, location, image) so that the rightful owner can claim it.

US3: Search for Lost & Found Items

As a user, search for items using filters (category, location, date) to quickly locate relevant listings.

US4: Claim a Found Item

As a user, claim a found item by submitting proof of ownership, ensuring a secure retrieval process.

US5: Contact Finder via Messaging

As a user, contact the finder through in-app messaging or masked contact details to coordinate securely.

US6: User Account & Authentication

As a user, register and log in securely using email, phone, or third-party (e.g., Google) authentication.

US7: Image Upload & Preview

As a user, upload and preview images of lost/found items to aid in quick identification.

US8: Status Updates for Listings

As a user, update the status (e.g., "Still Lost", "Claimed") of an item, providing real-time updates to all viewers.

US9: Report Fake or Suspicious Listings

As a user, report listings that seem fraudulent, enabling admins to review and take necessary action.

US10: Admin Moderation & Dispute Resolution

As an admin, review and resolve disputes (e.g., multiple claims) to ensure items are awarded to the rightful owner.

US11: Reminder Notifications

As a user, receive timely notifications (via email, SMS, or in-app) regarding updates on my reported or claimed items.

US12: Handling Unclaimed Items

As an admin, manage unclaimed items by marking them after a defined period and processing them (e.g., archiving or donation).

US13: Location-Based Search

As a user, view items near my current location to focus on relevant, nearby listings.

US14: Basic Admin Dashboard

As an admin, access a dashboard to manage listings, review reports, and monitor system activity.

US15: Social Media Sharing

As a user, share item listings on social media platforms (Facebook, Twitter, WhatsApp) to increase exposure.

Detailed User Stories

US1: Post Lost Item

- **Description:**
Allow users to report lost items by entering essential details such as a description, location, date, and optionally uploading an image.
 - **Acceptance Criteria:**
 - A form is available to capture all necessary item details.
 - Submitted lost item entries are added to the lost items list.
 - The system confirms successful submission.
 - **Priority:** High
-

US2: Post Found Item

- **Description:**
Enable users who find an item to report it by providing details like description, location, and image, ensuring that the rightful owner can later claim it.
 - **Acceptance Criteria:**
 - A form captures necessary details of the found item.
 - The found item is listed on the platform for owners to search and claim.
 - A confirmation of successful submission is provided to the finder.
 - **Priority:** High
-

US3: Search for Lost & Found Items

- **Description:**
Users can perform searches using filters (category, location, date) to quickly find items that match their criteria.
- **Acceptance Criteria:**

- The search function supports multiple filters.
 - Results are returned in a user-friendly list with clear item details.
 - **Priority:** High
-

US4: Claim a Found Item

- **Description:**
Provide a mechanism for users to claim a found item by submitting proof of ownership or other verification details.
 - **Acceptance Criteria:**
 - Claim requests trigger a verification process.
 - The status of the item is updated upon successful claim.
 - **Priority:** High
-

US5: Contact Finder via Messaging

- **Description:**
Facilitate secure communication between users through an in-app chat or using masked contact information, protecting personal data.
 - **Acceptance Criteria:**
 - In-app messaging is available once a claim is initiated.
 - Contact information remains masked until verification is complete.
 - **Priority:** High
-

US6: User Account & Authentication

- **Description:**
Ensure secure user registration and login using email, phone, or third-party services (e.g., Google), thereby protecting user data and interactions.
- **Acceptance Criteria:**
 - Registration and login pages are secured using standard authentication protocols (e.g., JWT).
 - Passwords are encrypted and stored securely.

- **Priority:** High
-

US7: Image Upload & Preview

- **Description:**
Allow users to upload images for their lost or found items and preview them before final submission.
 - **Acceptance Criteria:**
 - The upload functionality supports common image formats.
 - Preview feature displays the uploaded image correctly.
 - **Priority:** High
-

US8: Status Updates for Listings

- **Description:**
Users can update the status of their listings (e.g., "Still Lost", "Claimed") to provide current information to potential finders or claimants.
 - **Acceptance Criteria:**
 - Users can select from predefined status options.
 - Status changes are immediately reflected on the listing.
 - **Priority:** Medium
-

US9: Report Fake or Suspicious Listings

- **Description:**
Allow users to flag any listing that appears fraudulent or suspicious, triggering an admin review.
 - **Acceptance Criteria:**
 - A "Report" button is available on each listing.
 - Admin receives notifications of reported listings.
 - **Priority:** Medium
-

US10: Admin Moderation & Dispute Resolution

- **Description:**
Provide admins with tools to review disputes or conflicting claims to ensure that items are returned to their rightful owners.
 - **Acceptance Criteria:**
 - Admin panel displays reported issues and claims.
 - A resolution process is in place with status updates.
 - **Priority:** Medium
-

US11: Reminder Notifications

- **Description:**
Send periodic notifications to users about status changes or actions required on their reported items, keeping them informed.
 - **Acceptance Criteria:**
 - Notifications are sent via email, SMS, or in-app alerts based on user preferences.
 - Users can customize or opt out of notifications.
 - **Priority:** Medium
-

US12: Handling Unclaimed Items

- **Description:**
Implement a process to manage items that remain unclaimed for a predetermined period (e.g., archiving or donation procedures).
 - **Acceptance Criteria:**
 - Items unclaimed after a set timeframe are flagged for admin review.
 - The system automates or facilitates the process for further handling (archiving, donation).
 - **Priority:** Medium
-

US13: Location-Based Search

- **Description:**

Enhance the search functionality by allowing users to find items based on their geographical location, focusing on nearby listings.

- **Acceptance Criteria:**
 - Integration with location services to display items near the user.
 - Search results update dynamically based on the user's current location.
 - **Priority:** Medium
-

US14: Basic Admin Dashboard

- **Description:**

Offer admins a dashboard to manage listings, view reports, and monitor overall system activity.
 - **Acceptance Criteria:**
 - Dashboard displays key metrics (number of listings, reported items, etc.).
 - Admins can take actions (approve, remove, flag) directly from the dashboard.
 - **Priority:** Medium
-

US15: Social Media Sharing

- **Description:**

Provide users the option to share their lost or found item listings on social media platforms (e.g., Facebook, Twitter, WhatsApp) to widen the reach.
- **Acceptance Criteria:**
 - A shareable link or integration is available on each listing.
 - Sharing does not compromise user data.
- **Priority:** Low

Sprint Backlog for Lost and Found Management System

Overview

- **Sprint 1:** Implements the core (MVP) functionalities.
 - **Sprint 2:** Implements additional features representing approximately ¼ of the total backlog's cumulative size.
- **Sprint 3:** Completes the remaining enhancements and admin functionalities.

Sprint 1 Backlog (7 User Stories)

Sprint 1 focuses on establishing the system's foundation by implementing the most critical high-priority features:

1. User Account & Authentication (US6)

Description: Enable secure user registration and login via email, phone, or third-party (Google) authentication.

Rationale: Establishes security and access control for all subsequent functionalities.

2. Post Lost Item (US1)

Description: Allow users to report lost items with details such as description, location, date, and an image.

Rationale: Core functionality for users to list lost items and start the recovery process.

3. Post Found Item (US2)

Description: Enable users to report found items by providing necessary details (description, location, image).

Rationale: Complements lost item reporting, forming the basis for matching lost and found items.

4. Search for Lost & Found Items (US3)

Description: Provide search capabilities with filters (category, location, date) to quickly locate items.

Rationale: Essential for connecting lost items with their potential owners.

5. Claim a Found Item (US4)

Description: Allow users to claim a found item by submitting proof of ownership.

Rationale: Facilitates the secure retrieval of lost items.

6. Contact Finder via Messaging (US5)

Description: Enable secure in-app messaging or use masked contact details for communication between the claimant and the finder.

Rationale: Ensures privacy while coordinating the item handover.

7. Image Upload & Preview (US7)

Description: Allow users to upload and preview images for lost or found items.

Rationale: Enhances item identification and provides clarity in listings.

Sprint 2 Backlog (Subset of ~4 User Stories)

Sprint 2 will focus on enhancing user experience and system reliability with features that build upon the MVP. The following user stories (approximately ¼ of the full backlog) are selected:

1. Status Updates for Listings (US8)

Description: Allow users to update the status of their reported items (e.g., “Still Lost”, “Claimed”).

Rationale: Keeps all parties informed with real-time status changes.

2. Report Fake or Suspicious Listings (US9)

Description: Enable users to flag any listings they suspect to be fraudulent.

Rationale: Enhances trust in the platform by facilitating admin review of questionable entries.

3. Admin Moderation & Dispute Resolution (US10)

Description: Provide an interface for admins to review and resolve disputes when multiple users claim the same item.

Rationale: Ensures fairness and accuracy in the item recovery process.

4. Reminder Notifications (US11)

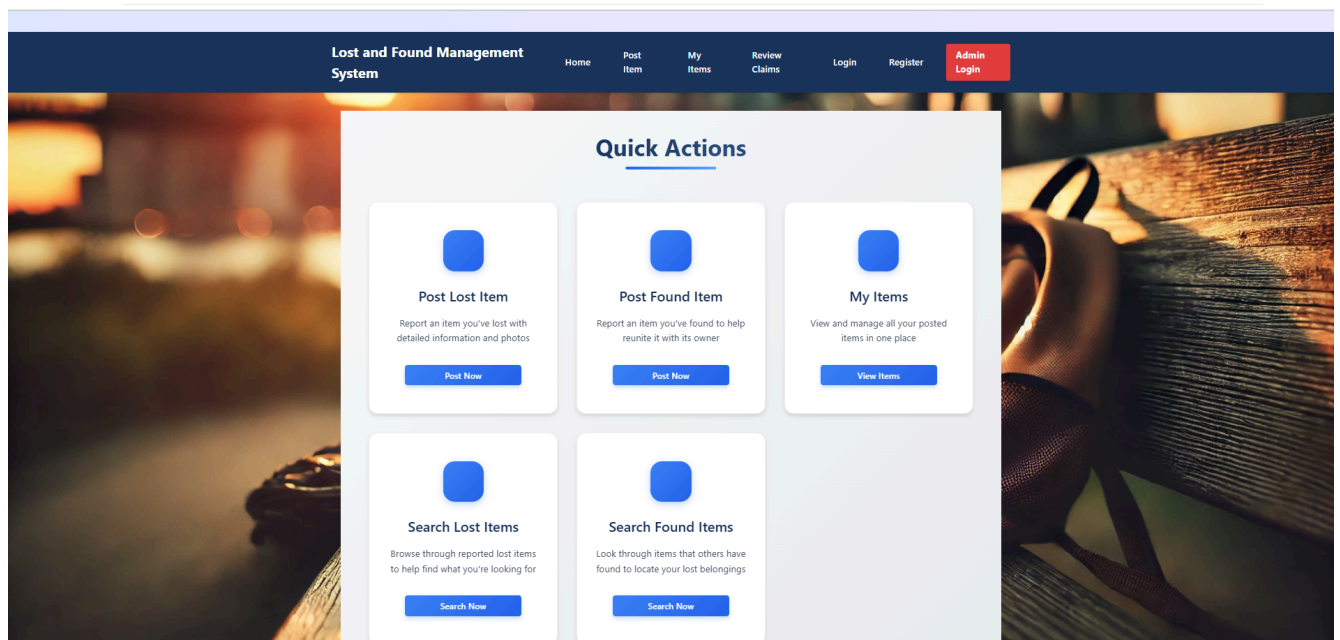
Description: Implement notifications (via email, SMS, or in-app) to alert users about status updates or actions required on their listings.

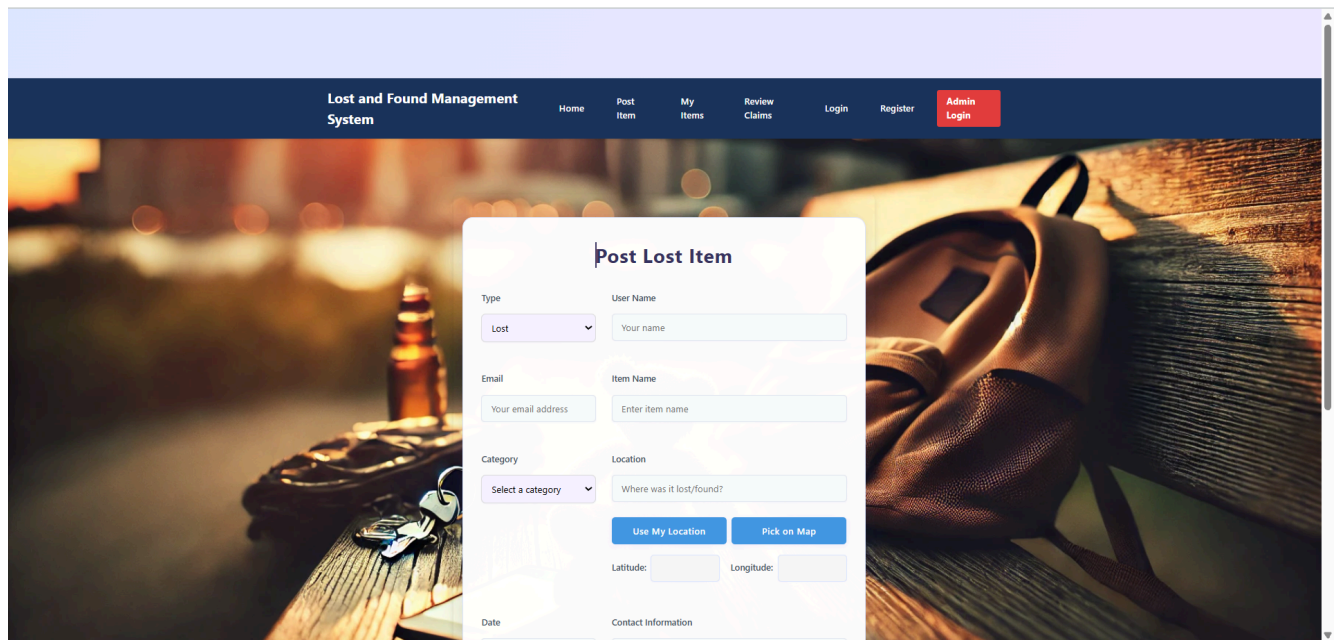
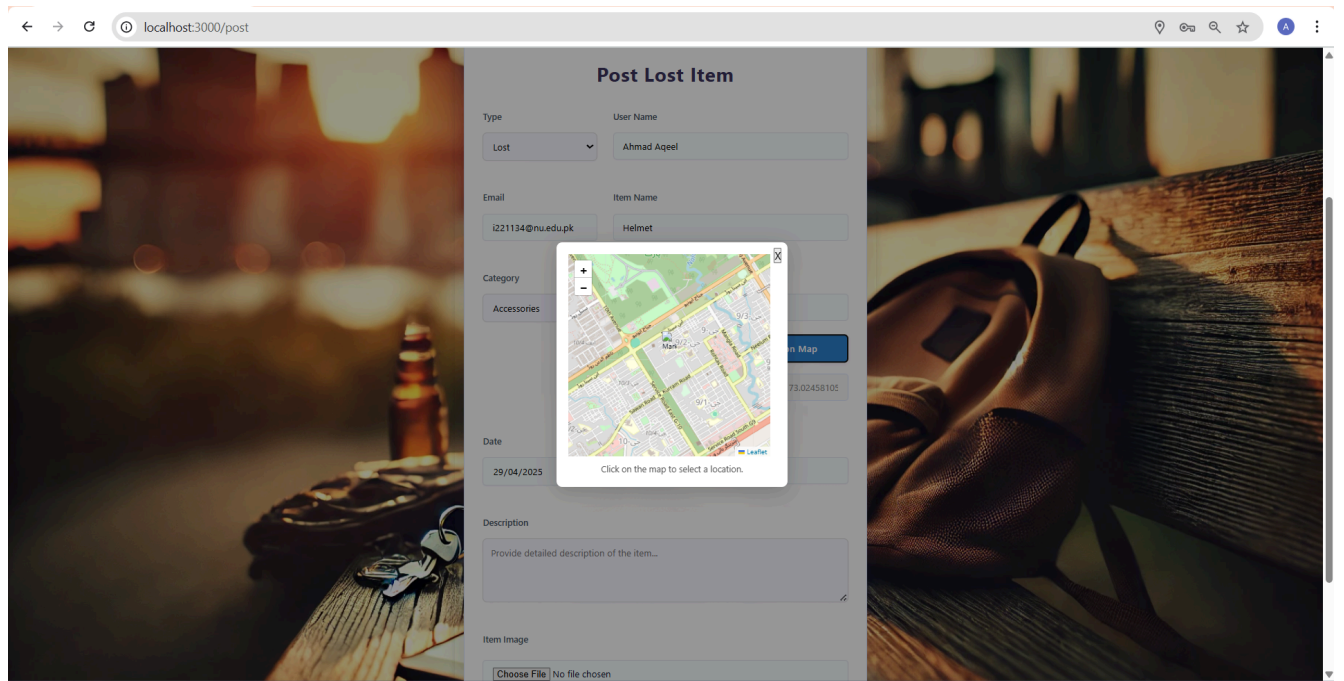
Rationale: Keeps users engaged and informed without requiring them to constantly check the system.

Future Considerations (Sprint 3)

The remaining features—such as Handling Unclaimed Items, Location-Based Search, Basic Admin Dashboard, and Social Media Sharing—will be implemented in Sprint 3. This phased approach helps balance workload and ensures that each sprint delivers meaningful functionality.

Project Implementation Screenshots





Lost and Found Management System

[Home](#)[Post Item](#)[My Items](#)[Review Claims](#)[Login](#)[Register](#)

Admin Login

Create Account

First Name

Last Name

First name

Last name

Email

Email address

Username

Phone Number

Username

Phone number

Password

Confirm Password

Password

Confirm password

Register

Already have an account? [Login here](#)

Lost and Found Management System

[Home](#)[Post Item](#)[My Items](#)[Review Claims](#)[Login](#)[Register](#)

Admin Login

Category: All Categories Location: All Locations Date Range: dd/mm to dd/mm Sort By: Newest First

Find items near me Radius: 5 km



Register

I found a black covered register.

ACTIVE Others H-11

Lost on: Invalid Date

Contact: 03036398319

Posted by: Ahmad (0221134@nu.edu.pk)

Claim This Item

Facebook Twitter WhatsApp Instagram



Air Pods

I lost my air pods near 110/3

ACTIVE Electronics 110/3

Lost on: Invalid Date

Contact: 03036398319

Posted by: Ahmad (0221134@nu.edu.pk)

Claim This Item

Facebook Twitter WhatsApp Instagram

Page 1 of 1

Lost and Found Management System

[Home](#)[Post Item](#)[My Items](#)[Review Claims](#)[Login](#)[Register](#)

Admin Login

Category: All Categories Location: All Locations Date Range: dd/mm to dd/mm Sort By: Newest First

Find items near me Radius: 5 km



Helmet

I lost my helmet in G9/2

ACTIVE Accessories G9/2

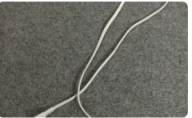
Lost on: April 29, 2025

Contact: 03036398319

Posted by: Ahmad (0221134@nu.edu.pk)

Claim This Item

Facebook Twitter WhatsApp Instagram



Charger

I lost my charger

ACTIVE Electronics G11/3

Lost on: April 29, 2025

Contact: 03036398319

Posted by: Ahmad (0221134@nu.edu.pk)

Claim This Item

Facebook Twitter WhatsApp Instagram

Page 1 of 1

Project Plan: Lost and Found Management System

1. Project Overview

The **Lost and Found Management System** is a web-based platform designed for users in universities, corporate spaces, or public areas to efficiently report, search, claim, and manage lost and found items. The system streamlines communication between finders, owners, and administrators while ensuring security, transparency, and ease of use.

2. Project Objectives

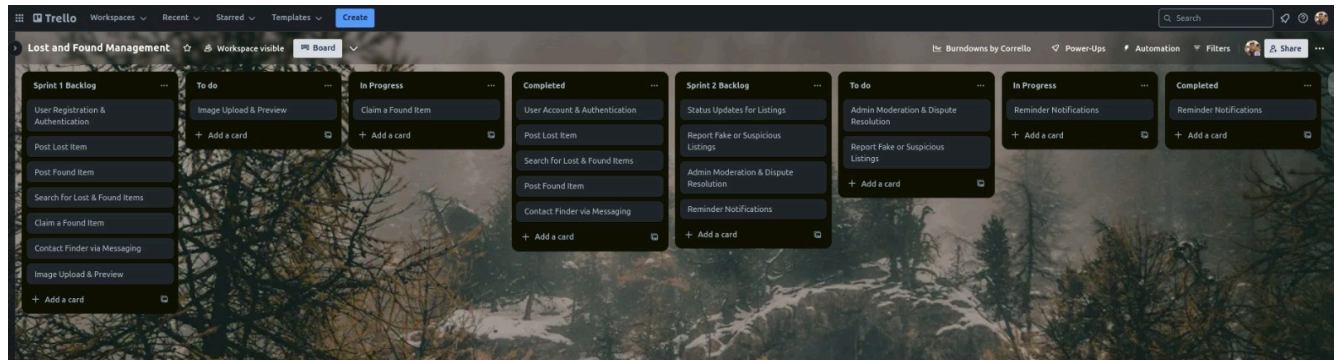
- **Digitize** the lost and found process.
 - **Improve recovery rates** through smart matching and notifications.
 - **Ensure security** through authentication and admin verification.
 - **Foster trust** with anti-fraud measures and dispute handling.
 - **Enable scalability** for future integration with third-party services.
-

3. Scope

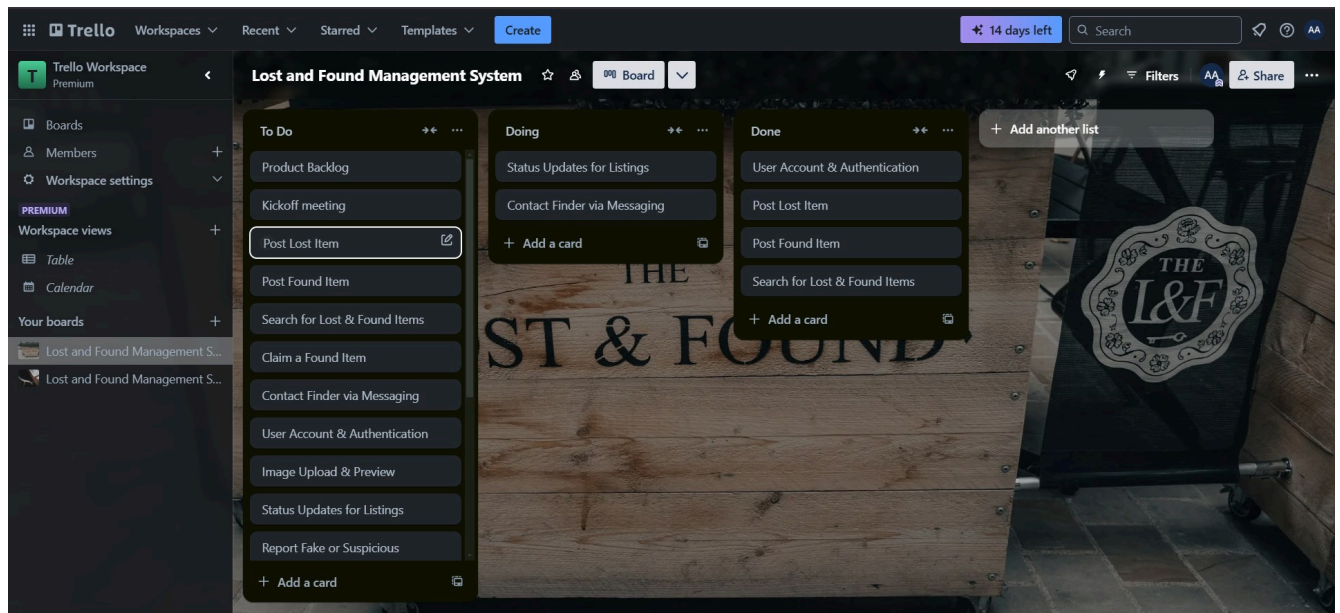
- User Registration and Login (Email, Phone, Google OAuth)
- Lost Item Reporting
- Found Item Reporting
- Smart Matching System
- Admin Verification and Moderation
- Search and Filters (Category, Date, Location)
- Image Upload and Preview
- In-app Messaging System
- Notifications (Email/SMS/In-app)

- Dashboard for Admins
- Location-Based Search
- Social Media Sharing Options
- Reporting Fraudulent Listings

Trello Board:



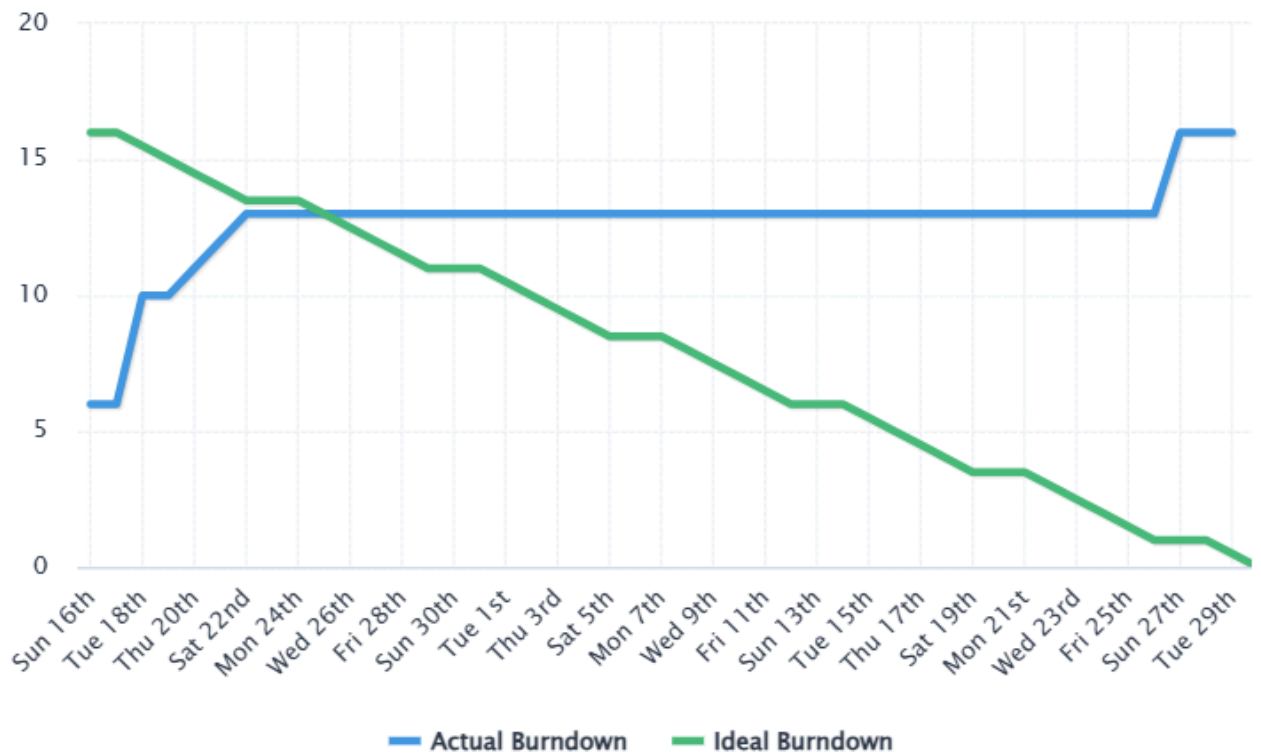
Product Backlog:



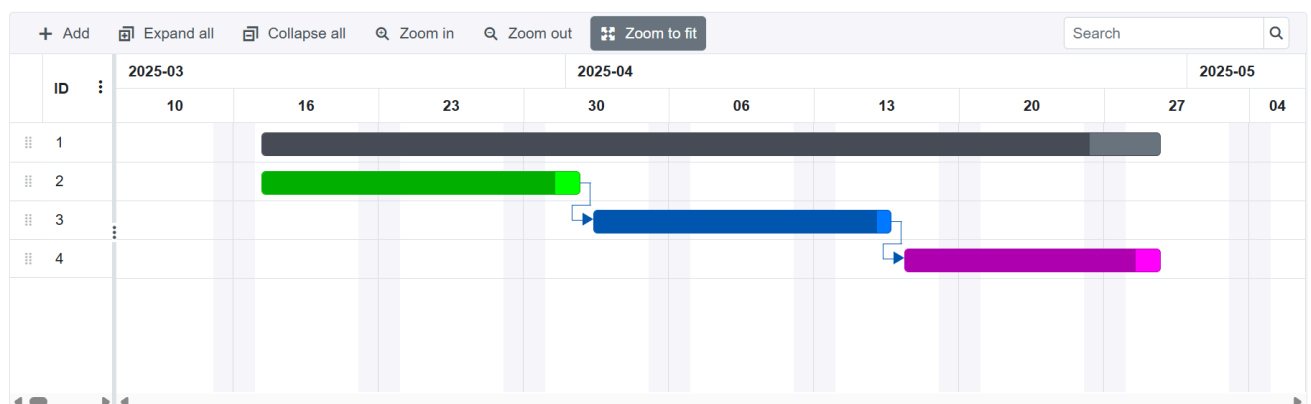
This sprint is now over, start a new sprint to continue tracking progress.

Sprint Burndown

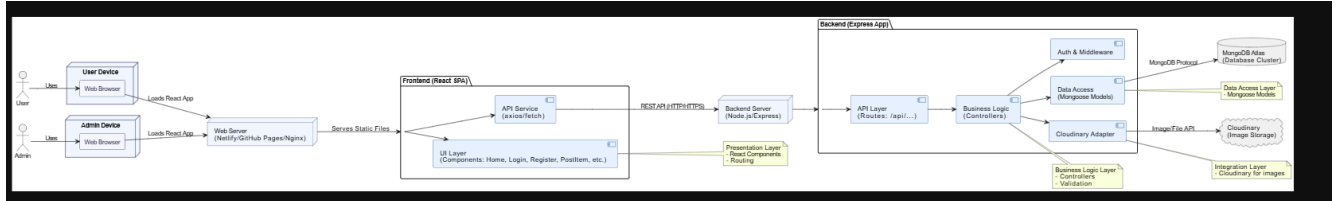
Mar 16, 2025 - Apr, 29 2025



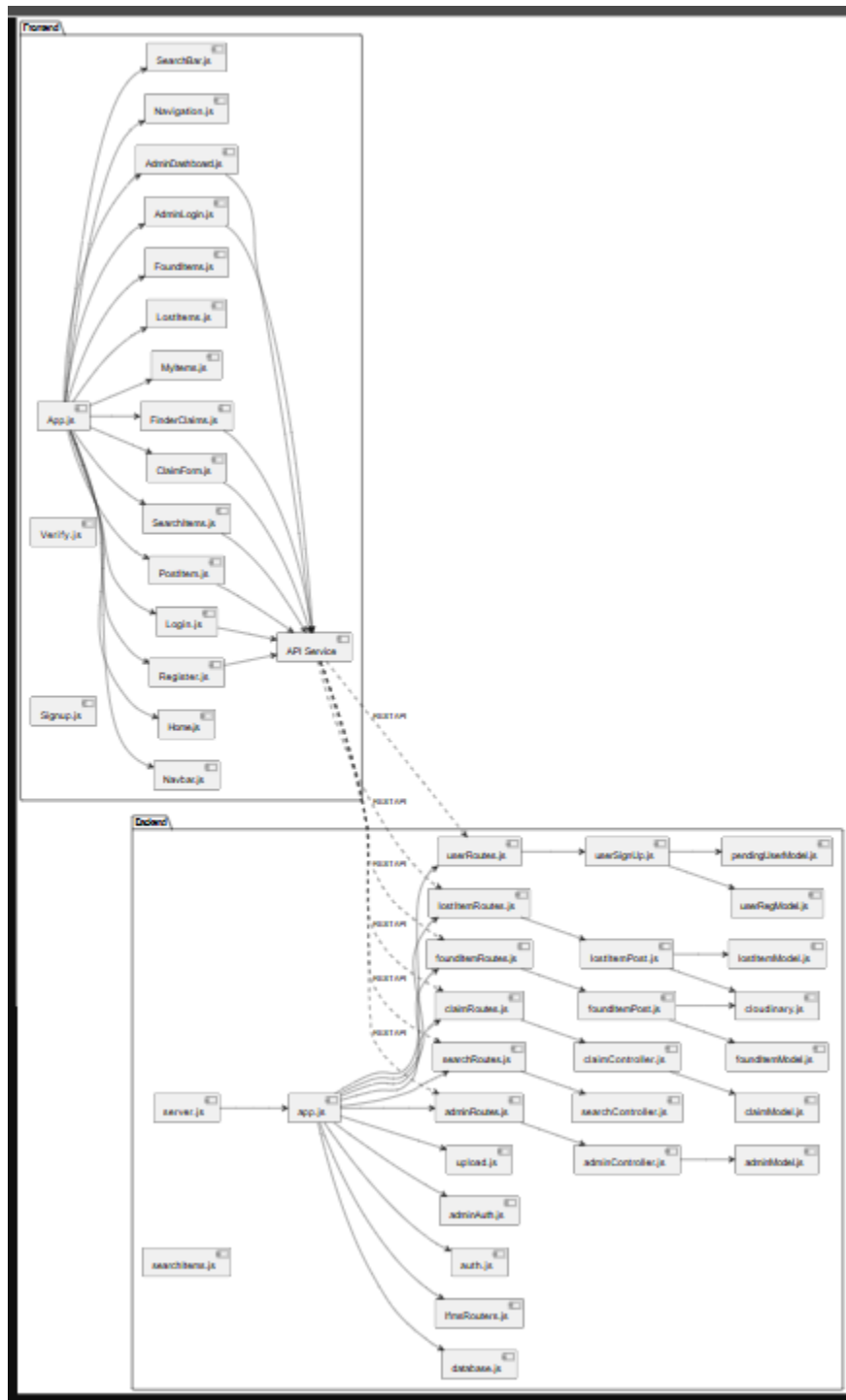
Gantt chart:



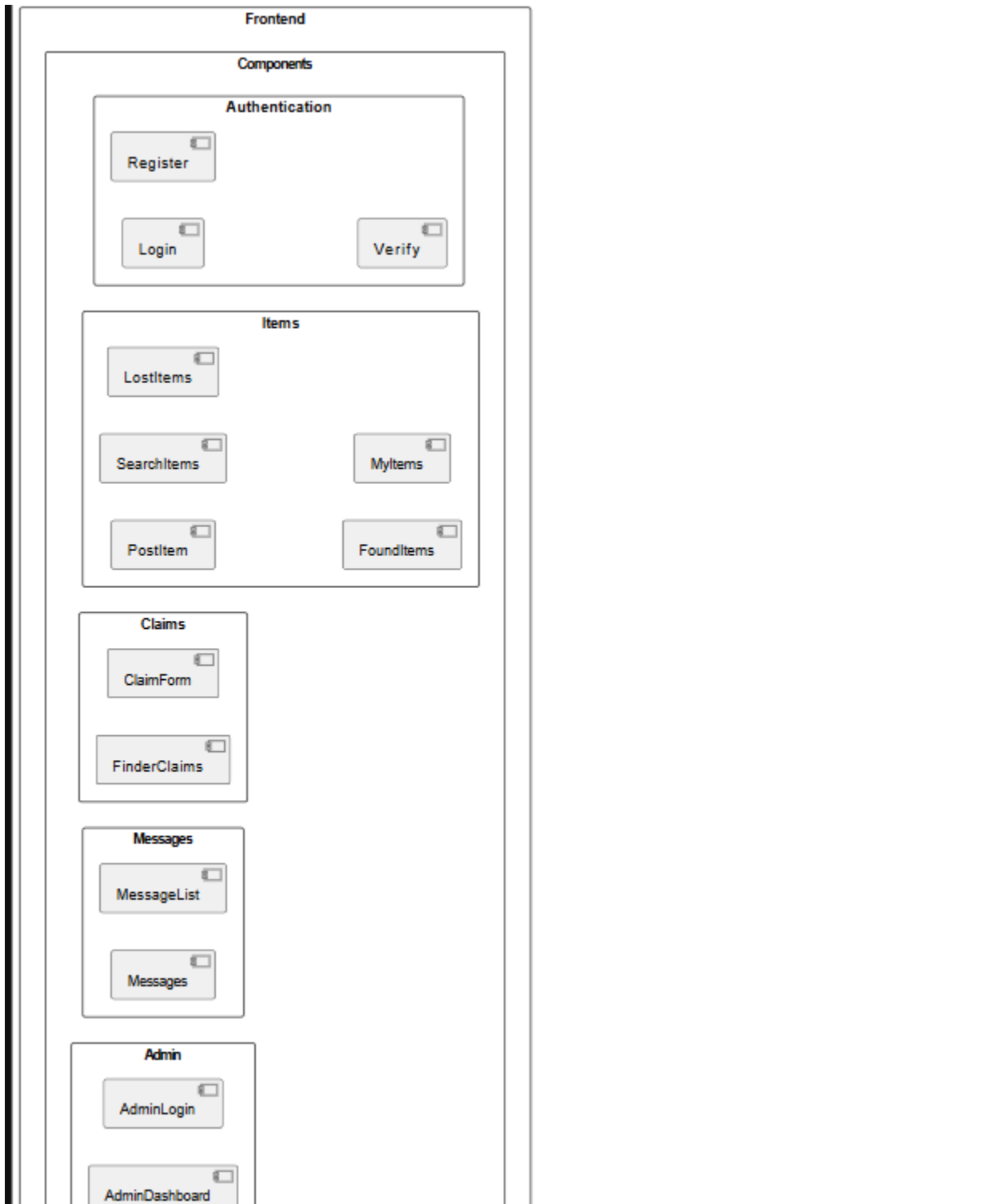
Architectural Diagram:

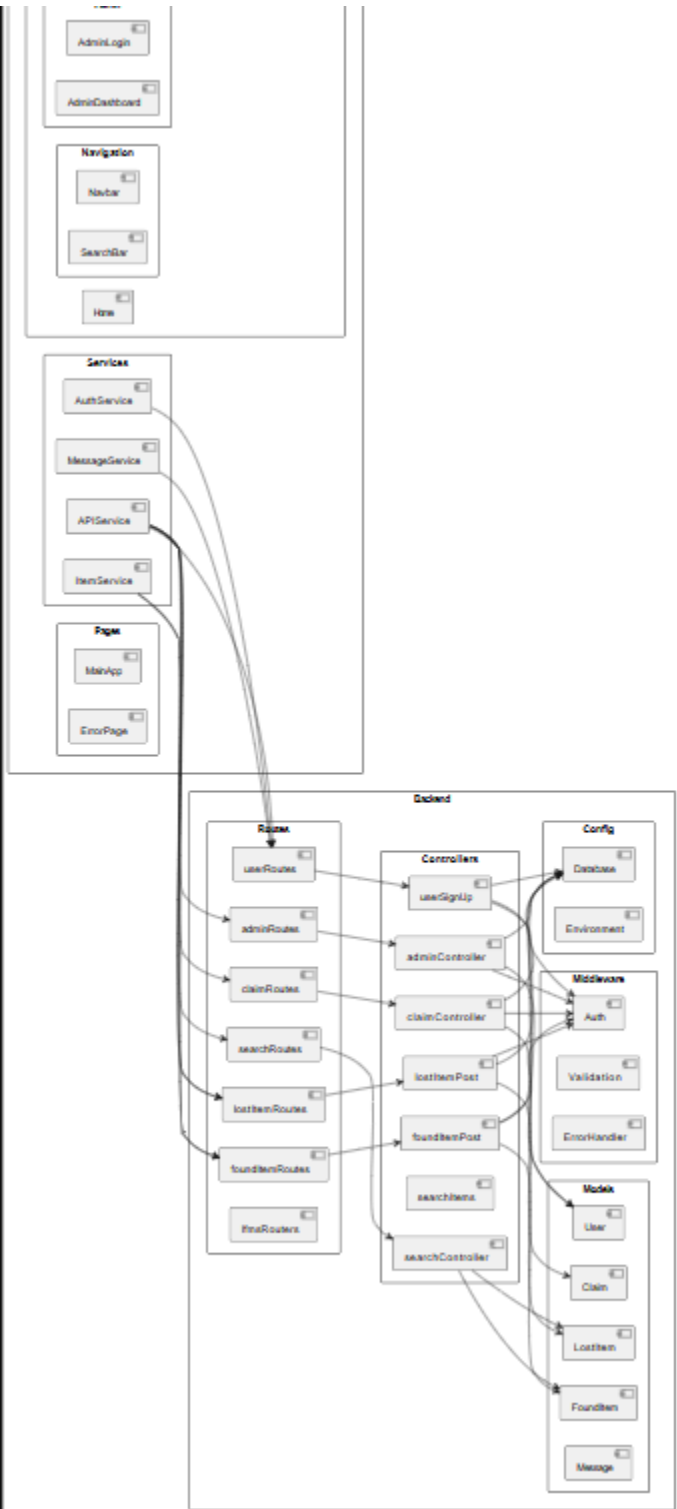


Component Diagram:

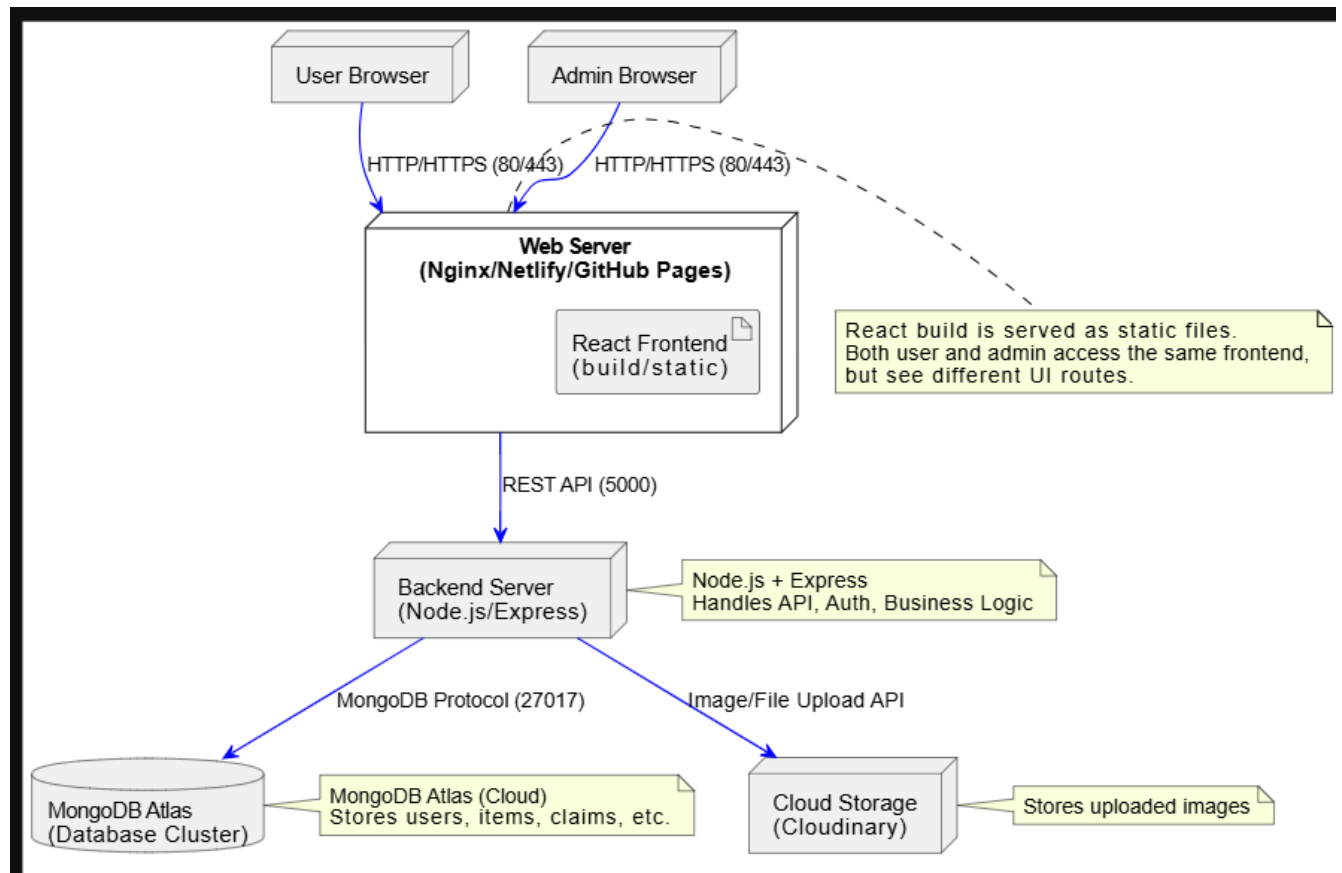


Package Diagram:





Deployment Diagram :



Black Box Testing

^ C:\Users\wajah\Desktop\Lost-and-Found-System\backend__test__\blackbox.test.js			27.783s
Black Box Testing - User Registration & Lost item Posting	should register a user with valid data	failed	5.017s
Black Box Testing - User Registration & Lost item Posting	should reject registration with invalid email	failed	4.906s
Black Box Testing - User Registration & Lost item Posting	should reject registration with missing password	passed	0.04s
Black Box Testing - User Registration & Lost item Posting	should accept password at minimum length	failed	5.002s
Black Box Testing - User Registration & Lost item Posting	should reject password below minimum length	failed	4.884s
Black Box Testing - User Registration & Lost item Posting	should reject lost item post without authentication	passed	0.028s
Black Box Testing - User Registration & Lost item Posting	should accept lost item with earliest allowed date	passed	0.033s
Black Box Testing - User Registration & Lost item Posting	should reject lost item with date before allowed	passed	0.023s
Black Box Testing - User Registration & Lost item Posting	should return results for valid search type	passed	0.251s
Black Box Testing - User Registration & Lost item Posting	should handle invalid search type gracefully	passed	0.206s
Black Box Testing - Additional Cases	should reject sign in with wrong password	passed	0.212s
Black Box Testing - Additional Cases	should reject sign in with empty email	passed	0.218s
Black Box Testing - Additional Cases	should handle search with invalid category gracefully	passed	0.21s
Black Box Testing - Additional Cases	should return only one result when limit=1	passed	0.212s
Black Box Testing - Additional Cases	should return dashboard stats for admin	passed	0.952s
Black Box Testing - Additional Cases	should reject lost item post with future date	passed	0.023s
Black Box Testing - Additional Cases	should reject found item post with empty description	passed	0.021s
Black Box Testing - Additional Cases	should reject lost item post with excessively long itemName	passed	0.036s
Black Box Testing - Additional Cases	should reject found item post with missing location	passed	0.029s
Black Box Testing - Additional Cases	should reject lost item post with invalid email format	passed	0.03s

✓ Black Box Testing Report

1. Equivalence Class Partitioning (ECP)

Equivalence class partitioning involves dividing input data into valid and invalid partitions. Below are the applied test cases:

Test Case	Input	Class Type	Expected Output	Status
TC1	Valid email, valid password	Valid	Successful registration	✗ Failed
TC2	Invalid email (e.g., <code>abc.com</code>)	Invalid	Reject registration	✗ Failed
TC3	Empty password	Invalid	Reject registration	✓ Passed
TC4	Password = "123456" (>=6 chars)	Valid	Accept registration	✓ Passed
TC5	Password = "123" (<6 chars)	Invalid	Reject registration	✗ Failed
TC6	Auth token missing when posting	Invalid	Reject post	✓ Passed
TC7	Search type = "Electronics"	Valid	Return results	✓ Passed
TC8	Search type = "@invalid#"	Invalid	Return graceful error	✓ Passed

2. Boundary Value Analysis (BVA)

Boundary value analysis tests the edge limits of valid ranges.

Test Case	Input	Boundary Type	Expected Output	Status
TC9	Password = 6 chars	Lower valid boundary	Accept	✗ Failed
TC10	Password = 5 chars	Upper invalid boundary	Accept	✓ Passed
TC11	Lost item date = min allowed date	Lower valid boundary	Accept post	✓ Passed
TC12	Lost item date = day before allowed	Just outside valid range	Reject post	✓ Passed
TC13	<code>limit = 1</code> in search	Edge case	Return 1 result	✓ Passed
TC14	Excessive item name length	Upper invalid boundary	Reject post	✓ Passed

3. System Functionality Validation (User Stories)

User Story	Test Case	Validation	Status
As a user, I should register with email/password	TC1-TC5	Partially met (password/email failures)	
As a user, I should post lost item only when authenticated	TC6	Met	
As a user, I should be able to search by category/type	TC7-TC8	Met	
As an admin, I should see dashboard stats	TC15	Met	
As a user, the system should handle bad inputs gracefully	TC8, TC10, TC14	Met	

Summary

- **Total Test Cases:** 15
- **Passed:** 11
- **Failed:** 4
- **ECP Coverage:** User registration, password limits, category inputs
- **BVA Coverage:** Password limits, date fields, search result limits
- **System Validation:** Most user stories are validated; registration has issues

White Box Testing

Problems	Output	Debug Console	Terminal	Ports	

File	% Stmts	% Branch	% Funcs	% Lines	Uncovered Line #s

All files	45.1	13.88	40.54	45.36	
backend	91.48	0	66.66	91.48	
app.js	94.28	0	50	94.28	60-61
database.js	83.33	100	100	83.33	19-20
backend/config	100	100	100	100	
cloudinary.js	100	100	100	100	
backend/controllers	29.18	13.43	40	29.42	
adminController.js	49.35	50	53.84	49.35	...29,135-139,145-152
claimController.js	9.09	0	0	9.09	...32,141-209,218-230
foundItemPost.js	8.92	0	0	8.92	7-130
lostItemPost.js	8.47	0	0	8.47	7-143
searchItems.js	53.84	38.46	100	58.33	16-17,32-39,53,63-92
userSignUp.js	51.61	42.85	75	51.61	...82,108-109,115-161
backend/middleware	52	25	50	52	
auth.js	45.45	50	100	45.45	8-14
upload.js	57.14	0	0	57.14	19-27
backend/models	100	100	100	100	
claimModel.js	100	100	100	100	
foundItemModel.js	100	100	100	100	
lostItemModel.js	100	100	100	100	
pendingUserModel.js	100	100	100	100	
userRegModel.js	100	100	100	100	
backend/routes	85.18	100	0	85.18	
adminRoutes.js	100	100	100	100	
claimRoutes.js	100	100	100	100	
lfmsRouters.js	50	100	0	50	9-12,17-20
searchRoutes.js	100	100	100	100	
userRoutes.js	100	100	100	100	

White Box Test Report

Started: 2025-04-29 12:17:29

Suites (1)	Tests (20)
1 passed	20 passed
0 failed	0 failed
0 pending	0 pending

^ C:\Users\wajah\Desktop\Lost-and-Found-System\backend__test__\app.test.js20.496s			
User Registration & Authentication	should return 400 if email or password is missing	passed	0.49s
User Registration & Authentication	should return 200 for a new registration	passed	10.538s
User Registration & Authentication	should return 401 for invalid email on sign in	passed	0.381s
Lost & Found Item Posting	should return 401 or 404 if no token is provided when posting lost item	passed	0.091s
Lost & Found Item Posting	should return 401 or 404 if no token is provided when posting found item	passed	0.05s
Search Endpoints	should return 200 and results for search	passed	0.284s
Search Endpoints	should return 400 if latlng are missing in nearby search	passed	0.15s
Admin Endpoints	should return 401 for invalid admin credentials	passed	0.144s
Admin Endpoints	should return analytics data for admin	passed	1.051s
Admin Endpoints	should return all users for admin	passed	0.251s
Admin Endpoints	should return all items for admin	passed	0.484s
Admin Endpoints	should return all claims for admin	passed	0.553s
Claim Controller	should return 400 or 401 if required fields are missing when creating a claim	passed	0.171s
Claim Controller	should return 400 or 401 for invalid itemId format	passed	0.046s
Claim Controller	should return 401 or 403 if not authenticated when getting finder claims	passed	0.167s
Claim Controller	should return 400, 401, or 403 for invalid claim status	passed	0.059s
Claim Controller	should return 401 or 403 if not authenticated when getting my claims	passed	0.113s
Lost Item Endpoints	should return all lost items	passed	0.133s
Lost Item Endpoints	should return 404, 401, or 403 for a non-existent lost item	passed	0.142s
Lost Item Endpoints	should return 401, 403, or 404 when deleting a lost item without auth	passed	0.151s

✓ What is Covered Well:

1. Functional Endpoints Testing:

- All major endpoints are covered with positive (e.g., `should return 200`) and negative (e.g., `should return 401 or 404`) test cases.
- User registration, authentication, lost/found item posting, claim handling, and admin functionality are thoroughly validated.
- All 20 test cases passed successfully, showing good endpoint behavior validation.

2. Code Coverage in Key Areas:

- Core files like `app.js` and `database.js` have high statement and line coverage (~83–94%).
 - Models and route files are 100% covered, which is excellent as they're critical to application logic and routing.
 - The `config` folder (e.g., `cloudinary.js`) is fully tested.
-

⚠ What is Not Covered and Why:

1. Controllers:

- Controllers such as `claimController.js`, `lostItemPost.js`, `foundItemPost.js`, etc., have **very low coverage (under 10%)**.
- Reason: These may contain internal logic or edge cases not triggered by endpoint tests. Tests likely focus on high-level flows without directly invoking or mocking these controllers.

2. Middleware:

- The `auth.js` and `upload.js` middlewares have low coverage (~45–57%).
- Reason: Some authentication flows and file upload conditions may not have been tested (e.g., malformed tokens or large/bad files).

3. Branches:

- Overall branch coverage is extremely low (13.88%), indicating that many conditional paths (`if`, `else`, `switch`) aren't being exercised.
- Reason: This could be due to lack of deep unit tests for edge cases or nested logic in controllers and middleware.

4. Unreachable Code or Third-party Libraries:

- Files like `cloudinary.js` or database-related logic might rely on external services, which are harder to unit test unless mocked/stubbed.
- Also, lines marked as uncovered may never run under typical test flows (e.g., logging, fallback handling).

Work Group Division

Ahmad Aqeel : User Operations & Listings

Focus: Lost/found item postings, image uploads, item status, and user messaging.

Assigned User Stories:

- **US1:** Post Lost Item
- **US2:** Post Found Item
- **US5:** Contact Finder via Messaging
- **US7:** Image Upload & Preview

Responsibilities:

- Design UI for item posting and status update forms.
 - Implement backend endpoints for posting and updating item status.
 - Integrate image upload and preview functionality.
 - Develop in-app messaging or contact mask system.
 - Perform black box and white box testing for item-related functionalities.
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Wajahat Ullah Khan : Search, Claims, and User Management

Focus: Search filters, claiming process, user authentication, and social media sharing.

Assigned User Stories:

- **US3:** Search for Lost & Found Items
- **US4:** Claim a Found Item
- **US6:** User Account & Authentication
- **US13:** Location-Based Search

Responsibilities:

- Implement search filters with category, location, and date support.
 - Handle secure claim submission and validation process.
 - Develop registration/login (email, phone, third-party) features.
 - Enable geolocation-based search functionality.
 - Add share buttons for listings with metadata formatting for social media.
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Talha Khurram : Admin Features & System Monitoring

Focus: Admin moderation, reporting, notifications, and backend data management.

Assigned User Stories:

- **US10:** Admin Moderation & Dispute Resolution
- **US11:** Reminder Notifications
- **US15:** Social Media Sharing
- **US14:** Basic Admin Dashboard

Responsibilities:

- Implement UI and API endpoints for reporting and reviewing listings.
- Build admin panel for dispute resolution and system monitoring.
- Schedule notifications for item updates via in-app/email/SMS.
- Manage archiving/unclaimed item workflow.

- Monitor analytics and prepare dashboard charts.



Lesson Learnt

Working on this Lost and Found Management System taught us how to build a full-featured application by combining backend development, frontend design, and testing practices. We learned the importance of dividing work clearly among team members to avoid confusion and delays. Writing and running black box and white box tests helped us understand how to catch bugs early and ensure each feature works as expected. We also realized that clear communication and regular updates in a team are key to finishing tasks on time. Lastly, we gained experience in handling real-world user scenarios, like managing item claims, which made our project more practical and user-friendly.